

Problem Set 3 (45 pts)

Due at 11:59pm

Friday, September 19

All the problems are from <https://aimacode.github.io/aima-exercises/> (possibly) with minor typo fixes, slight rewriting, or addition of missing information.

1. **(18 pts)** (Exercise 3.27) Trace the operation of A* search applied to the problem of getting to Bucharest from Lugoj using the straight-line distance heuristic. That is, show the sequence of nodes that the algorithm will consider and the f , g , and h scores for each node.
2. **(10 pts)** (Exercise 3.31) The heuristic path algorithm is a best-first search in which the evaluation function is $f(n) = (2 - w)g(n) + wh(n)$.
 - (a) **(3 pts)** For what values of w is this complete?
 - (b) **(4 pts)** For what values is it optimal, assuming that h is admissible?
 - (c) **(3 pts)** What kind of search does this perform for $w = 0$, $w = 1$, and $w = 2$?
3. **(17 pts)** (Exercise 3.33) There are n vehicles which occupy squares $(1, 1)$ through $(n, 1)$ (i.e., the leftmost column) of an $n \times n$ grid. The vehicles must be moved to the rightmost column but in reverse order; so the vehicle i that starts in $(i, 1)$ must end up in $(n - i + 1, n)$. On any time step, every one of the n vehicles can move one square up, down, left, or right, or stay put; but if a vehicle stays put, one other adjacent vehicle (but not more than one) can hop over it. No two vehicles can occupy the same square.
 - (a) **(4 pts)** What is the size of the state space as a function of n ?
 - (b) **(3 pts)** What is the branching factor as a function of n ?
 - (c) **(4 pts)** Suppose that vehicle i is at (x_i, y_i) ; write a nontrivial admissible heuristic h_i for the number of moves required to get the vehicle to its goal location $(n - i + 1, n)$, assuming no other vehicles are on the grid.
 - (d) **(6 pts)** Which of the following heuristics are admissible for the problem of moving all n vehicles to their destinations? Explain.
 1. $\sum_{i=1}^n h_i$.
 2. $\max\{h_1, \dots, h_n\}$.
 3. $\min\{h_1, \dots, h_n\}$.